

Yash Maini

2× IEEE Author | 500+ DSA Problems | AWS • MLflow • Terraform

+91-8920028757 | mainiyash2@gmail.com | yashmaini.tech

[LinkedIn](#) | [GitHub](#) | [LeetCode](#)

Experience

AIIMS Delhi | Medical Imaging Research

Jul 2025 – Sep 2025

Research Collaborator – DICOM Automation Service

New Delhi

- Developed automated video-to-DICOM pipeline using **Python**, **PostgreSQL**, **Docker**, **Orthanc PACS** under **Dr. Deepak Agrawal**, achieving **95% automation** and **30s processing/video**.

IBM SkillsBuild | Agentic AI: From Learner to Builder

Jul 2025 – Aug 2025

AI Agent Architect Training Program

Remote

- Completed 4-week training on **Agentic AI** and **AI Agent Architecture** with **CSRBOX Foundation**, building production-ready AI agents using **Gemini API**.

GGSIU – USAR | Deep Learning Research

Jul 2024 – Jun 2025

Undergraduate Research Associate

New Delhi

- Led DL research under **Dr. Sanjay Kumar Singh** and **Dr. Pranshu Saxena**; authored 2 IEEE papers on image classification using **transfer learning** and **Bayesian optimization**.
- Achieved **97.23%** accuracy in tomato disease detection (**Xception**) and **98.46%** in breast tumor classification (CNNs with **Population-Based Training**).

M.K. Associates | Python, Excel, ML, EDA

Jan 2024 – Jun 2024

Financial Data Analyst Intern

Part-Time (Remote)

- Analyzed **10K+ transactions**, identified 150+ anomalies across **700+ accounts**, and automated workflows (**40% efficiency gain**).
- Built ML models for risk classification achieving **93% accuracy**, improving audit precision and compliance.

Technical Skills

Languages: Python, SQL, Bash, Java, JavaScript

Libraries & Frameworks: Scikit-learn, TensorFlow, Keras, NumPy, Pandas, Matplotlib, Seaborn

MLOps & Tools: DVC, MLflow, Docker, Git, GitHub Actions, Flask, FastAPI, Terraform, Grafana, DagsHub

Databases: MySQL, PostgreSQL, SQLite, MongoDB

Cloud (AWS): EC2, S3, Lambda, ECS, ECR, RDS, Bedrock, SageMaker, CloudWatch, IAM, CLI, CDK

Platforms: Linux, Windows, macOS

Education

University School of Automation and Robotics (GGSIU)

Nov 2022 – May 2026 (Expected)

Bachelor of Technology – AI & Machine Learning; **GPA: 8.61** — till 6th semester

New Delhi, India

- Courses:** Artificial Intelligence, Machine Learning, Networking, Databases, Operating Systems, Data Structures, Analysis of Algorithms

Projects

CopyGuard – Serverless AI Code Detection Platform

[\[Code\]](#) [\[Blog\]](#) [\[Demo\]](#)

- Technologies:** AWS Lambda, Bedrock Claude v2, Terraform, S3, CloudFront, Grafana, MLOps
- Engineered a production-grade platform to **detect AI-generated code** using Amazon Bedrock (Claude v2), achieving **<2s response time** and **~99.9% availability**.
- Orchestrated full **IaC stack** (**API Gateway**, **IAM**, **CORS**, **logging**) via **Terraform & S3 + CloudFront**.
- Monitored latency/confidence metrics** via **Grafana**; stored **outputs versioned in S3**.

ThreatMatrix – End-to-End MLOps Pipeline for Network Intrusion Detection

[\[Code\]](#) [\[Blog\]](#) [\[Demo\]](#)

- Technologies:** Python, ML, FastAPI, MongoDB, MLflow, DagsHub, Docker, GitHub Actions, AWS EC2
- Built modular ML pipeline** with data ingestion, validation, training, and prediction components organized under custom internal package architecture.
- Containerized pipeline using Docker**, automated CI/CD through GitHub Actions, and deployed to **Amazon ECR** for reproducible workflows.
- Launched FastAPI service on AWS EC2** with real-time endpoints achieving sub-15ms latency, tracked via **MLflow and DagsHub**.

Publications

- Y. Maini, S. K. Singh and P. Saxena**, "Xception for Tomato Leaf Disease Detection: Hyperparameter Tuning and Fine-tuning Approaches," **2024 ICAIQSA**, Nagpur, India, [\[DOI\]](#) [\[Code\]](#) [\[Blog\]](#)
- Y. Maini, S. K. Singh and P. Saxena**, "Breast Tumor Classification with Fine-Tuned Hyperparameter Training using Deep Learning Models," **2025 AI-Driven Smart Healthcare for Society 5.0**, Kolkata, India, [\[DOI\]](#) [\[Code\]](#) [\[Blog\]](#)

Achievements

- Ranked in the **top 11% globally on LeetCode**, with over **500 problems solved** across core DSA topics
- Qualified GATE 2025** (Data Science & AI) – Top 8% nationwide, reflecting strong CS and ML fundamentals
- Leadership Coordinator, Coding Society**, leading coding events and peer activities